Dense Layer

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$$\underset{\text{input}}{\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}} \begin{array}{c} \text{Neuron 1} \quad\quad\quad\quad \text{Neuron n} \\ \begin{bmatrix} w_{11} & & w_{1n} \\ w_{21} & \cdots\cdots & w_{2n} \\ w_{31} & & w_{3n} \end{bmatrix} \end{array} + \underset{\text{bias}}{\begin{bmatrix} b_1 & \cdots & b_n \end{bmatrix}} = \underset{\text{output}}{\begin{bmatrix} y_1 & \cdots & y_n \end{bmatrix}}$$

we need to calculate dL/dw. we are given dL/dy, so we need dy/dw

$dy/dw \rightarrow y = w_1 x_1 \ldots\ldots + w_n x_n + \text{bias}$

then $dy/dx_1 = X$, then dy/dx (ALL) is just X

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \begin{bmatrix} dL/dy_1 & \cdots\cdots & dL/dy_n \end{bmatrix} = \begin{bmatrix} dL/dw_{11} & & dL/dw_{1n} \\ \vdots & \cdots\cdots & \vdots \\ dL/dw_{n1} & & dL/dw_{nn} \end{bmatrix}$$

The gradient of each weight

we want to return dL/dx. we have dL/dy

so we need to calculate $dL/dy \cdot dy/dx$

which can be done with:

$dL/dy \cdot w^T$